Proper Motions from the DESI Legacy Imaging Surveys

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University of Waterloo / Perimeter Institute

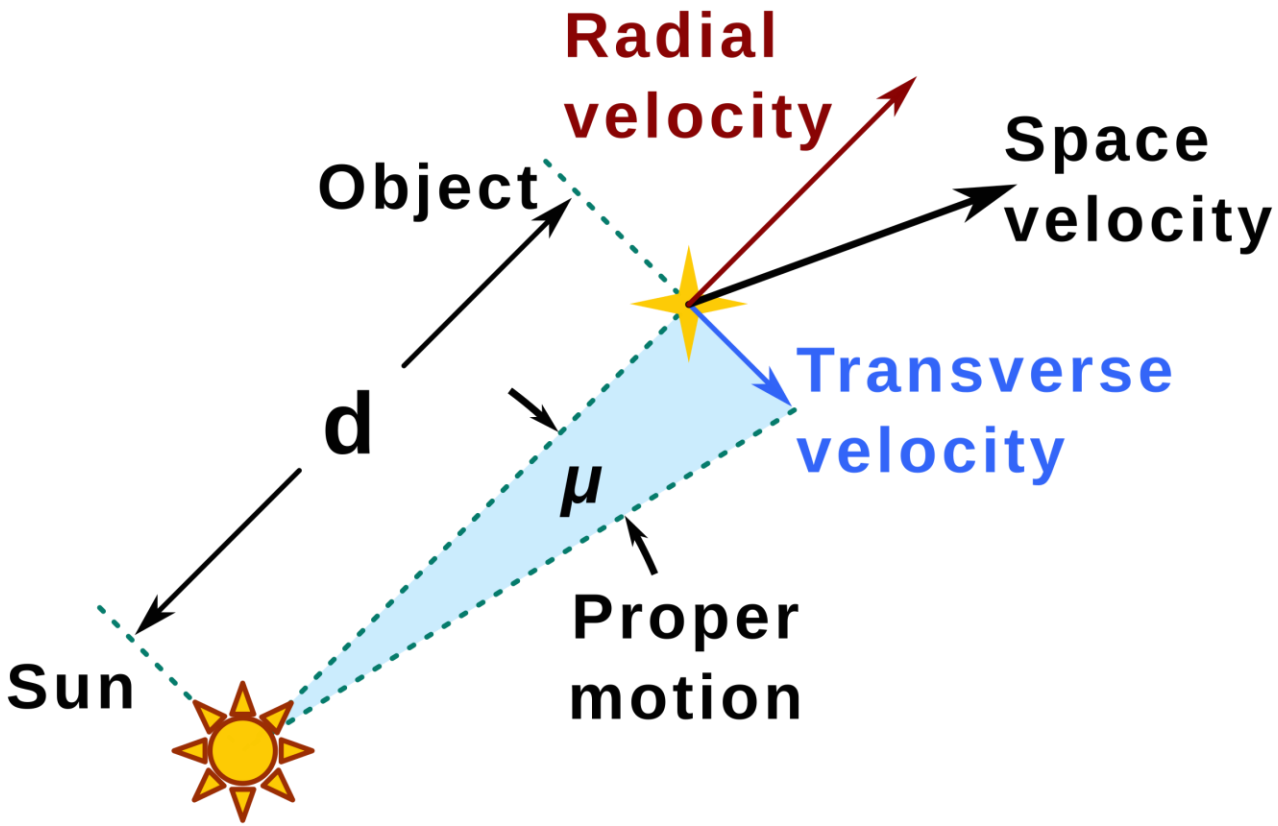
Advisors: Dustin Lang and James Taylor

Outline

- Why we need proper motions
- Gaia vs. DESI Legacy
- Differential Chromatic Refraction (DCR)
- CCD-specific corrections
- Lateral colour
- Results and error analysis
- Summary/To-do’s

- **Why we need proper motions**
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What proper motions are



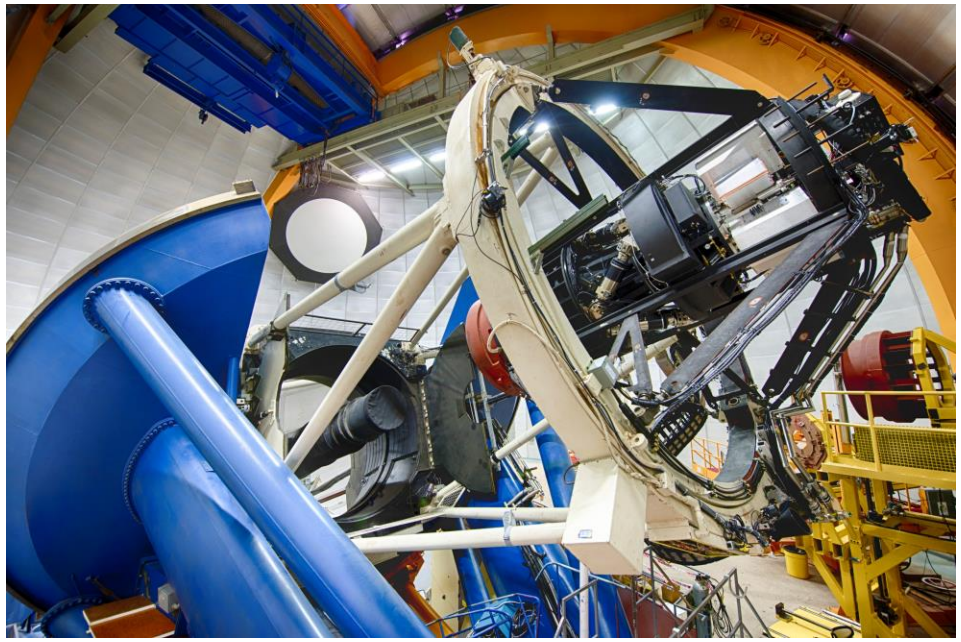
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Why we care about proper motions

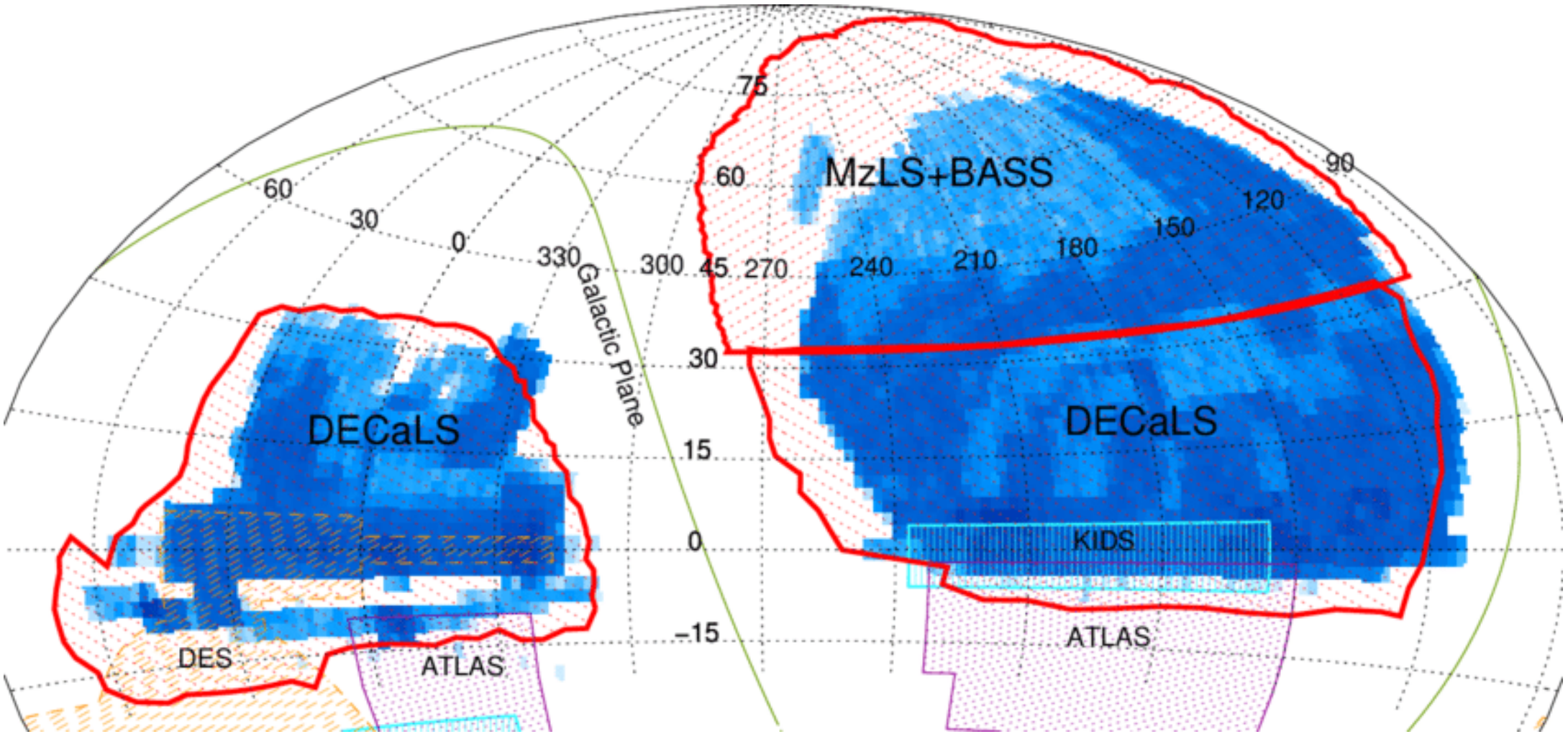
- Stellar dynamics and evolution
- Galactic structure and kinematics
- PM of galaxies and large-scale structure
- Find tidal tails
- Find fast stars

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The DESI Legacy Imaging Surveys

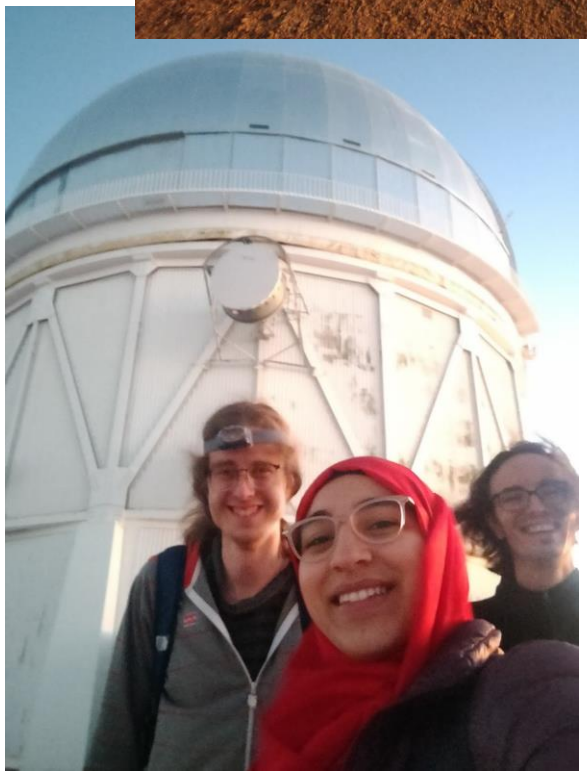


- Provide targets for the DESI spectroscopic survey, primarily distant galaxies
- Three collaborative initiatives: the **Dark Energy Camera Legacy Survey**, the Beijing-Arizona Sky Survey, and the Mayall z-band Legacy Survey
- Due to the combination of large field of view and high sensitivity from 400-1000 nm, DECam is an efficient option for obtaining photometry in the g, r, and z bands


MzLS+BASS
DECaLS
DECaLS
KIDS
ATLAS
ATLAS
DES
Galactic Plane
75
60
30
0
330
300
45
270
240
210
180
150
120
90
60
30
0
15
0
−15









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Gaia vs. DESI Legacy Imaging Surveys

Gaia

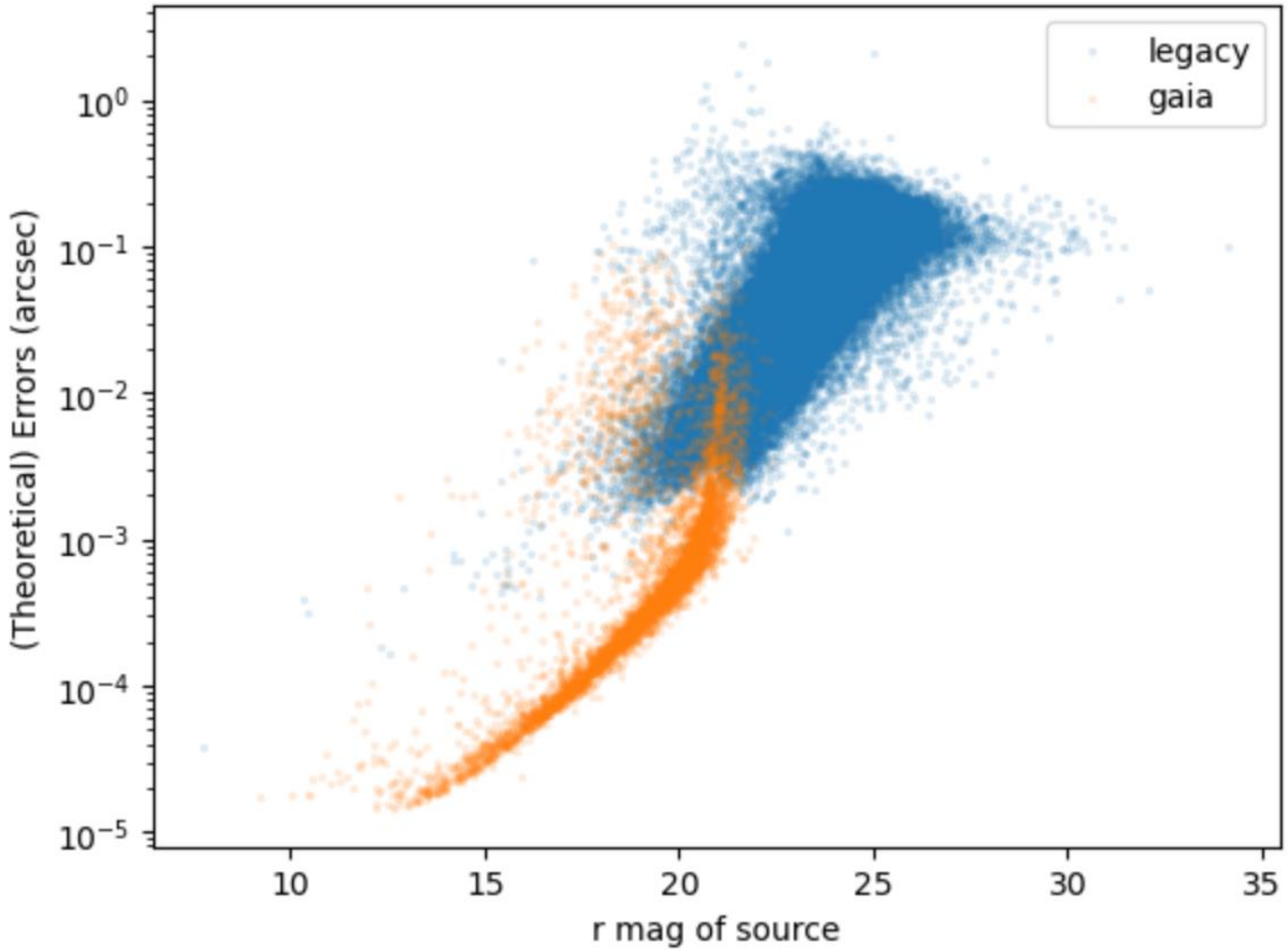
- Map the Milky Way
- All-sky
- Integrated broad-spectrum photometric instrument (BP – RP)
- 3-21 mags
- Error range 0.01-3 mas

Legacy (DECaLS/DECam)

- Map the large-scale universe
- 9,000 deg^2
- g, r, z-bands
- 14-27 mags

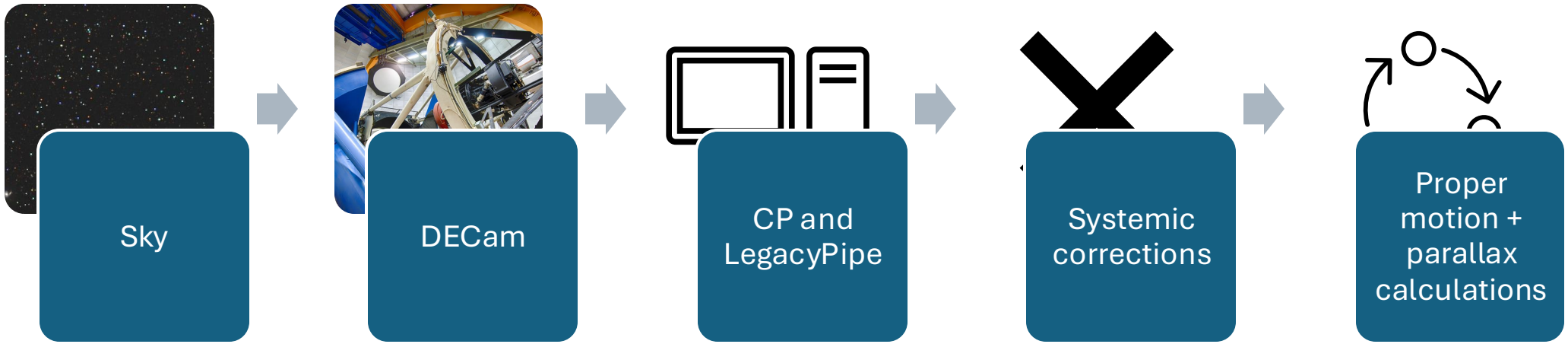
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The Overall Process



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DECam Community Pipeline

- Low-level calibration of images
- Instrumental signature removal

LegacyPipe

- Uses forced photometry
- Measures brightness and new position on stars

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LegacyPipe: Forced Photometry

Chat-GPT prompt: explain forced photometry to an elementary school kid

- Alright, let's imagine you have a picture of a lot of stars and galaxies in space, like a big space photograph. Now, some of these stars and galaxies are really bright and easy to see, like the Sun, while others are not so bright and might be hiding behind others.
- Forced photometry is like having a list of all the stars and galaxies you know should be in the picture. Even if you can't see some of them clearly because they're faint or hiding, you still want to know how much light (brightness) they have. So, instead of just looking at the whole picture and trying to find everything, you say, "Okay, I know these things are supposed to be here," and you focus on those specific spots.
- It's a bit like having a treasure map with X marks where the treasures are. Even if the treasures are buried or hard to spot, you go to those X marks and dig there to find out what's hidden. Similarly, forced photometry looks at the places where you expect certain stars or galaxies to be and figures out how bright they are, even if they're not shining very brightly in the picture.
- So, in simple terms, forced photometry helps astronomers find and measure the light of things in space, even if those things are a bit shy and try to hide in the cosmic crowd!

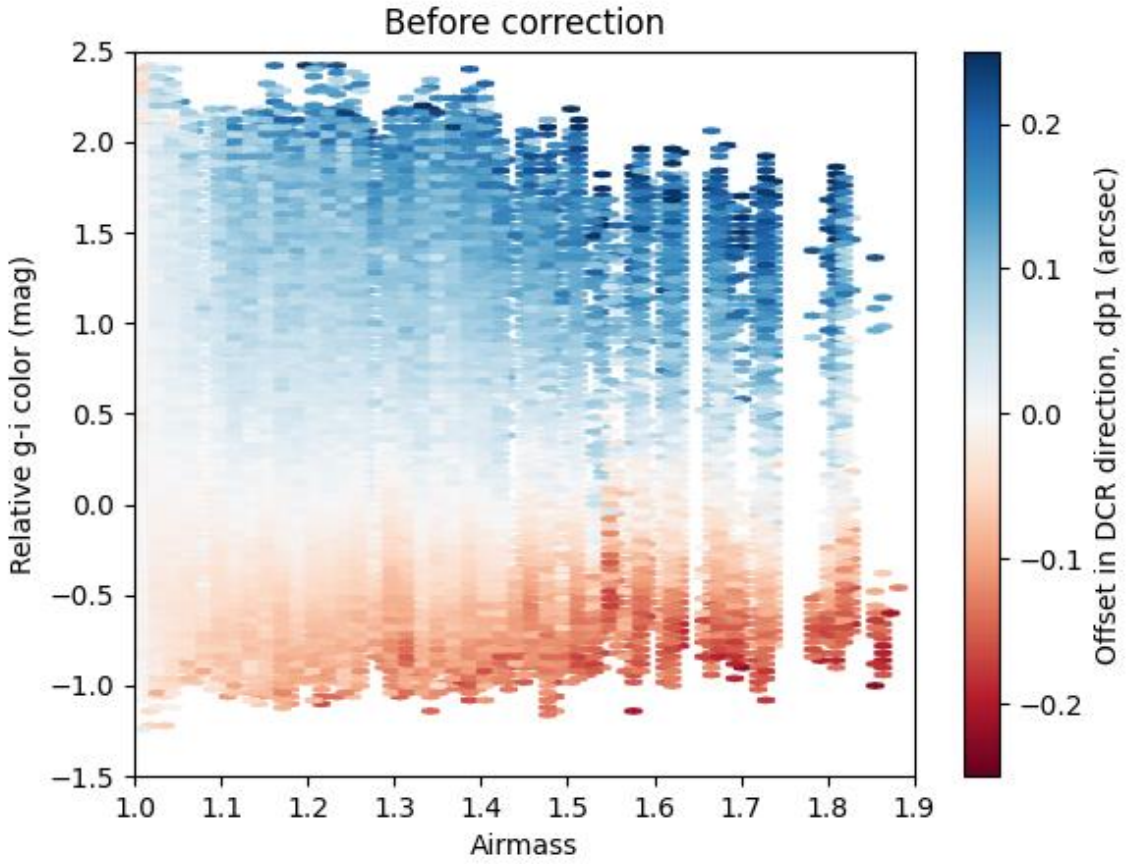
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Differential Chromatic Refraction (DCR)

$$\Delta x = K_b c \tan z \hat{\boldsymbol{p}}^*$$

- where c is the object color (g – i),
- $\hat{\boldsymbol{p}}$ is the unit sky vector toward the zenith (the parallactic angle),
- and K_b is a constant derivable for each band from the instrumental bandpass and atmospheric index of refraction

*Bernstein et. al. (2017)



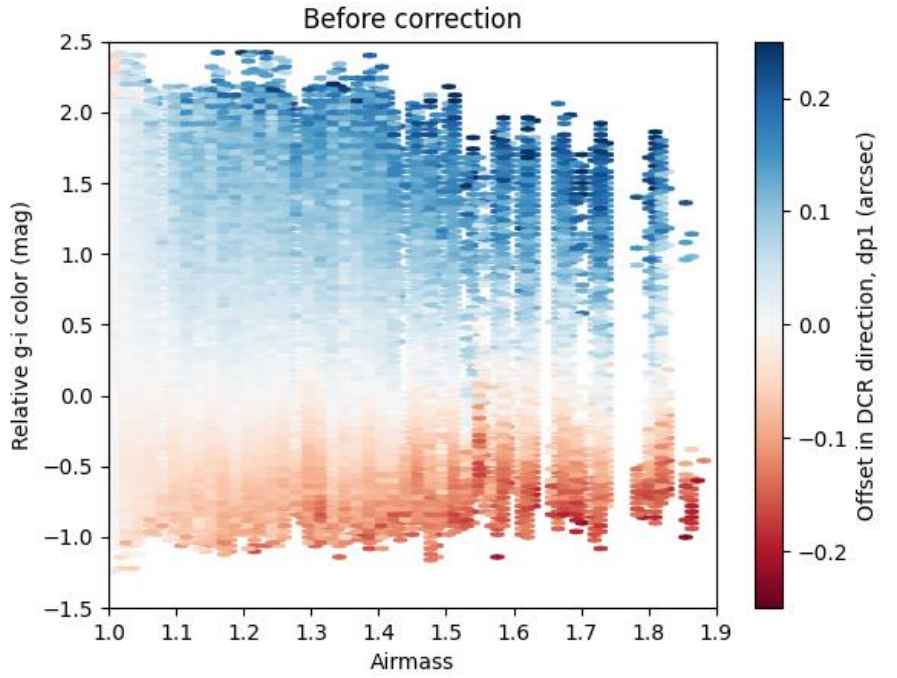
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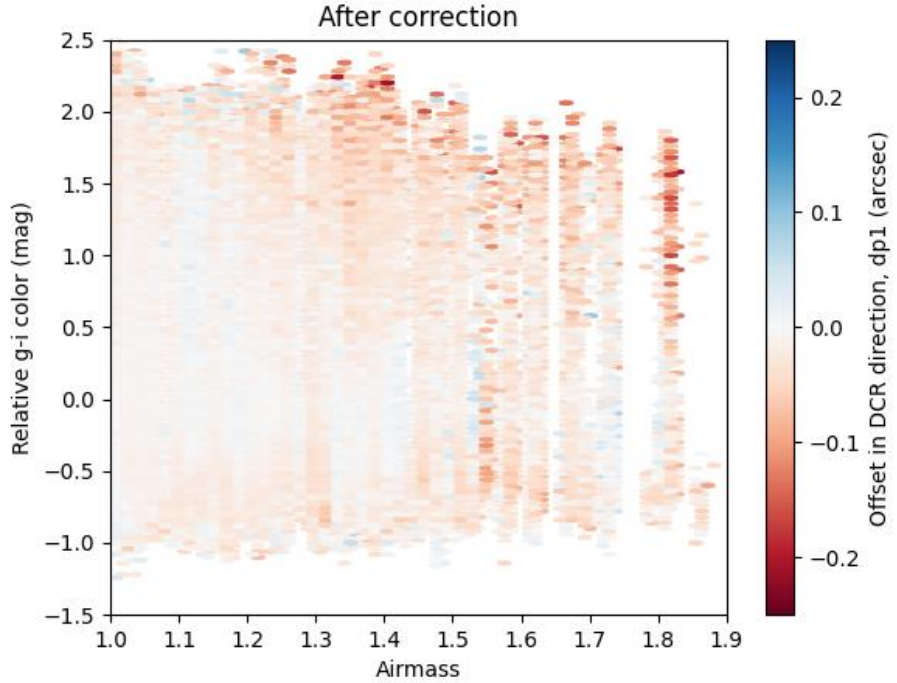
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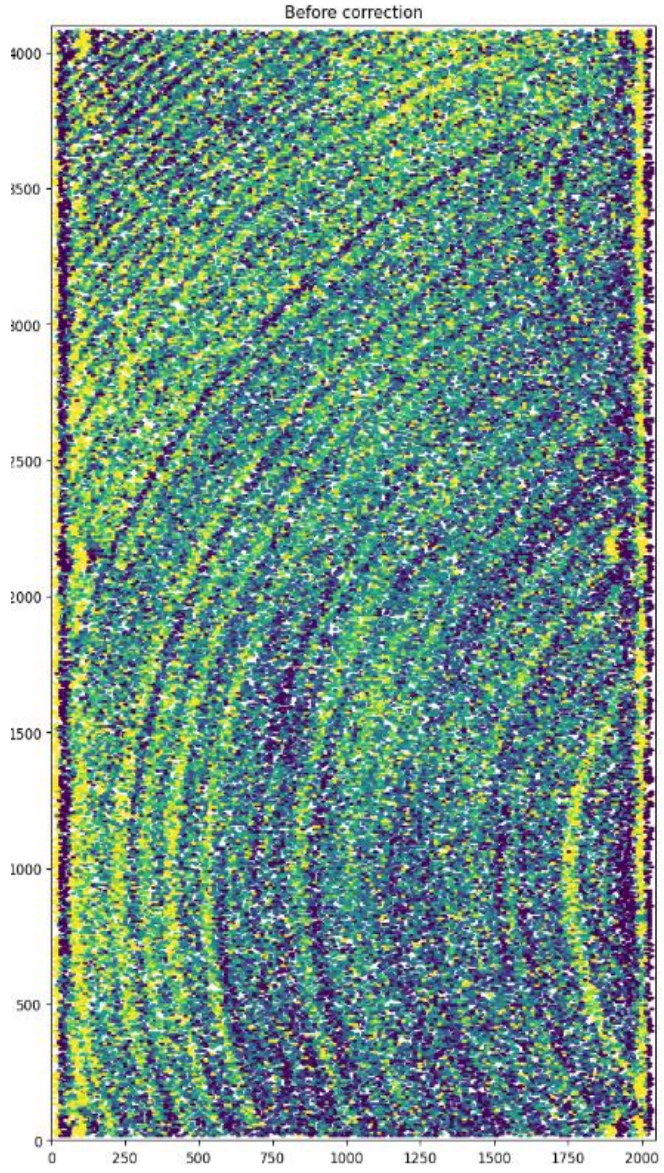
*full_fit_dr
measurem

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CCD (Charge-Coupled Device) Corrections

- “Tree rings” result from impurity density fluctuations in the silicon boules from which the CCD wafers are cut.

- Compare measured and known positions of Gaia stars
- Create these “distortion maps”
- Apply to other stars in the catalogue

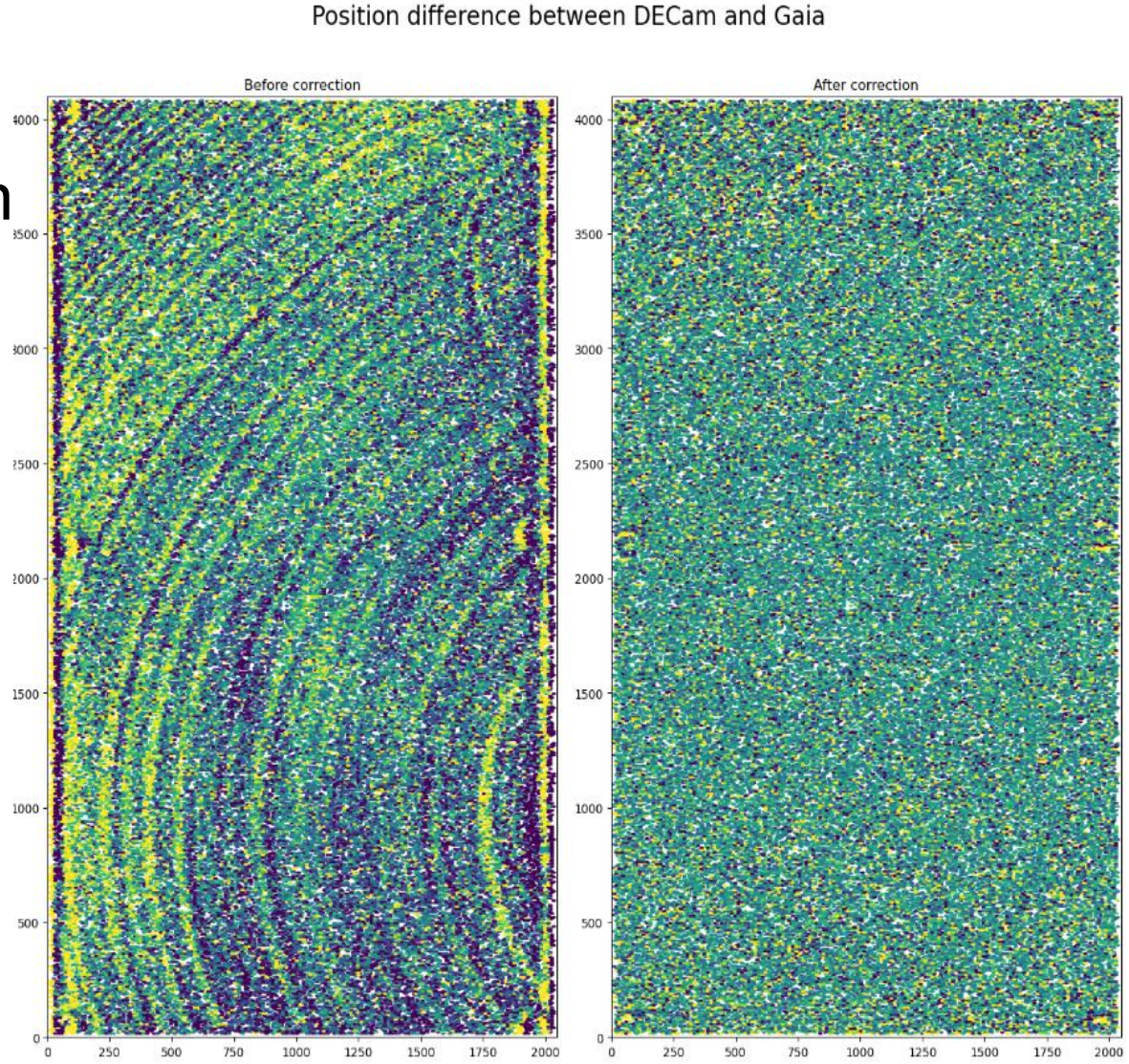


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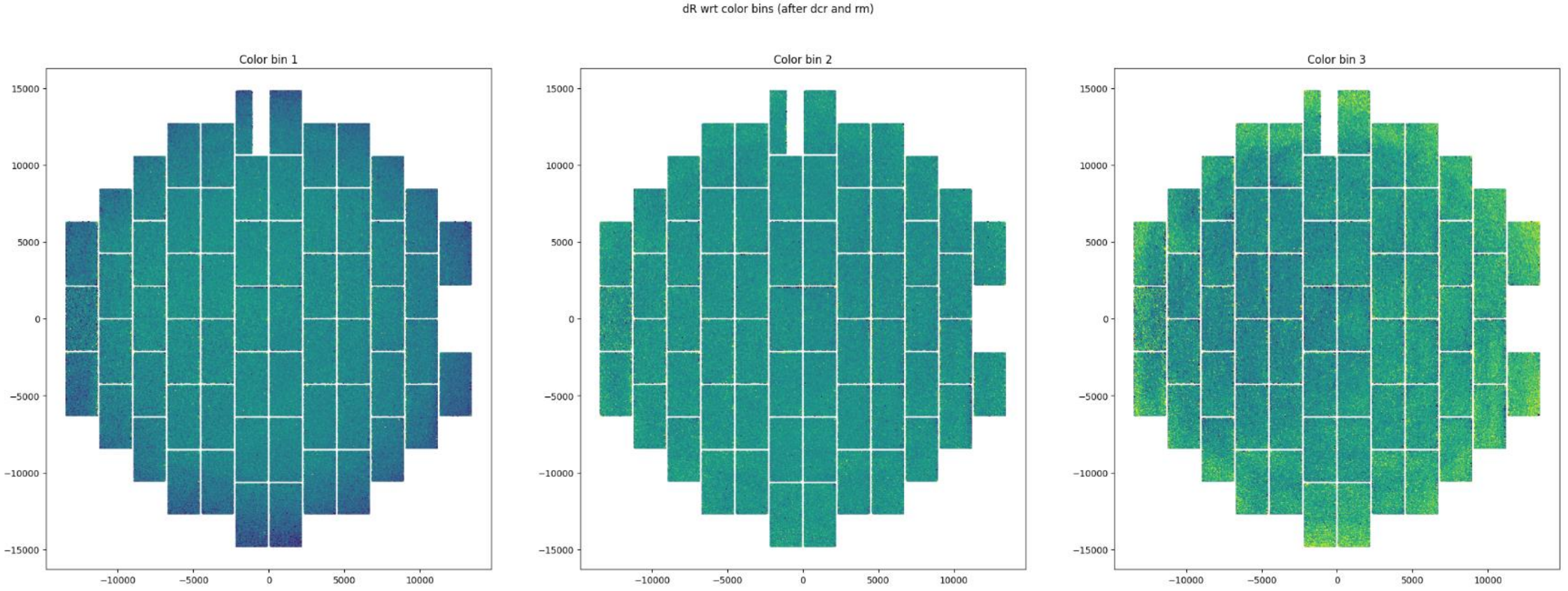
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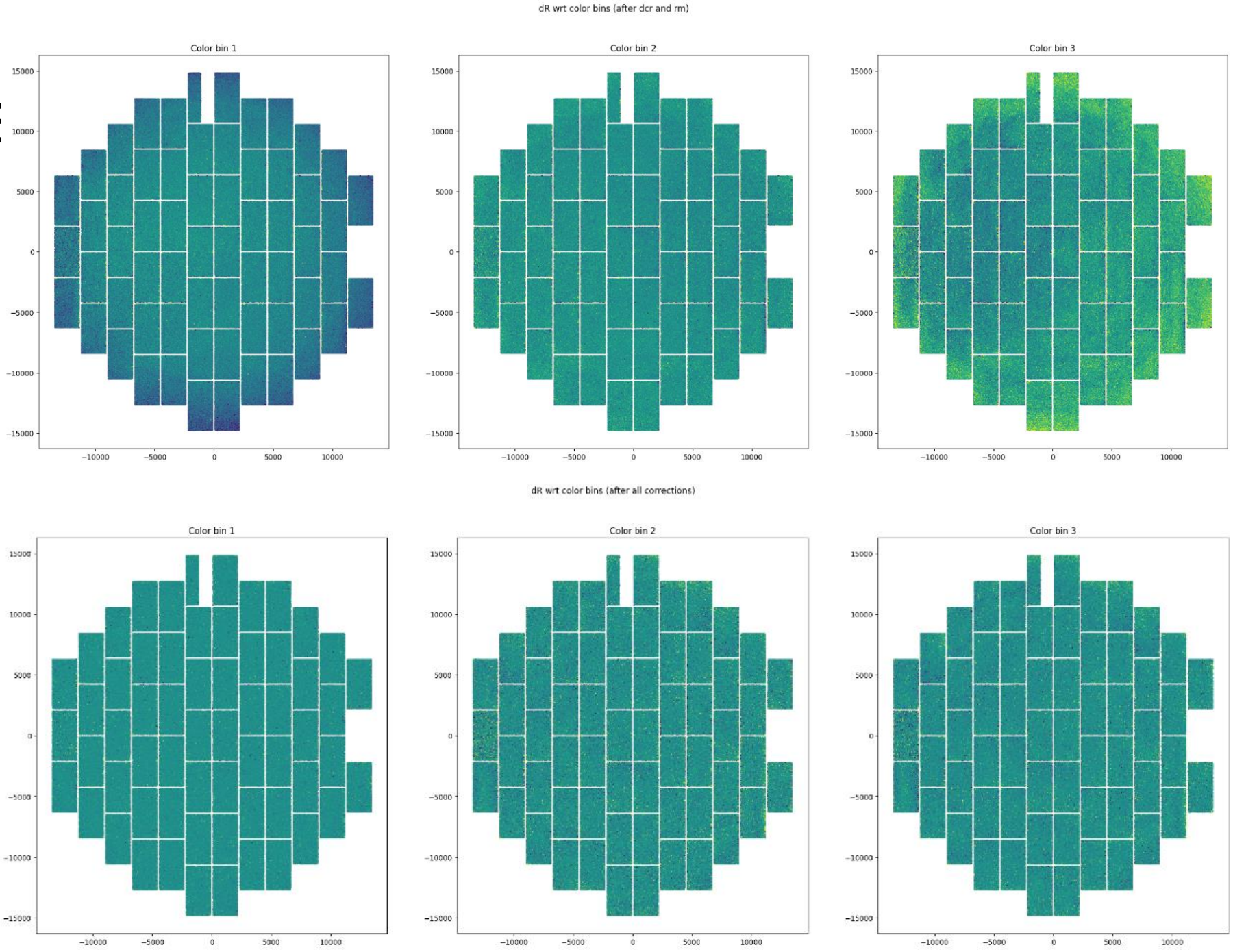
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Lateral Colour



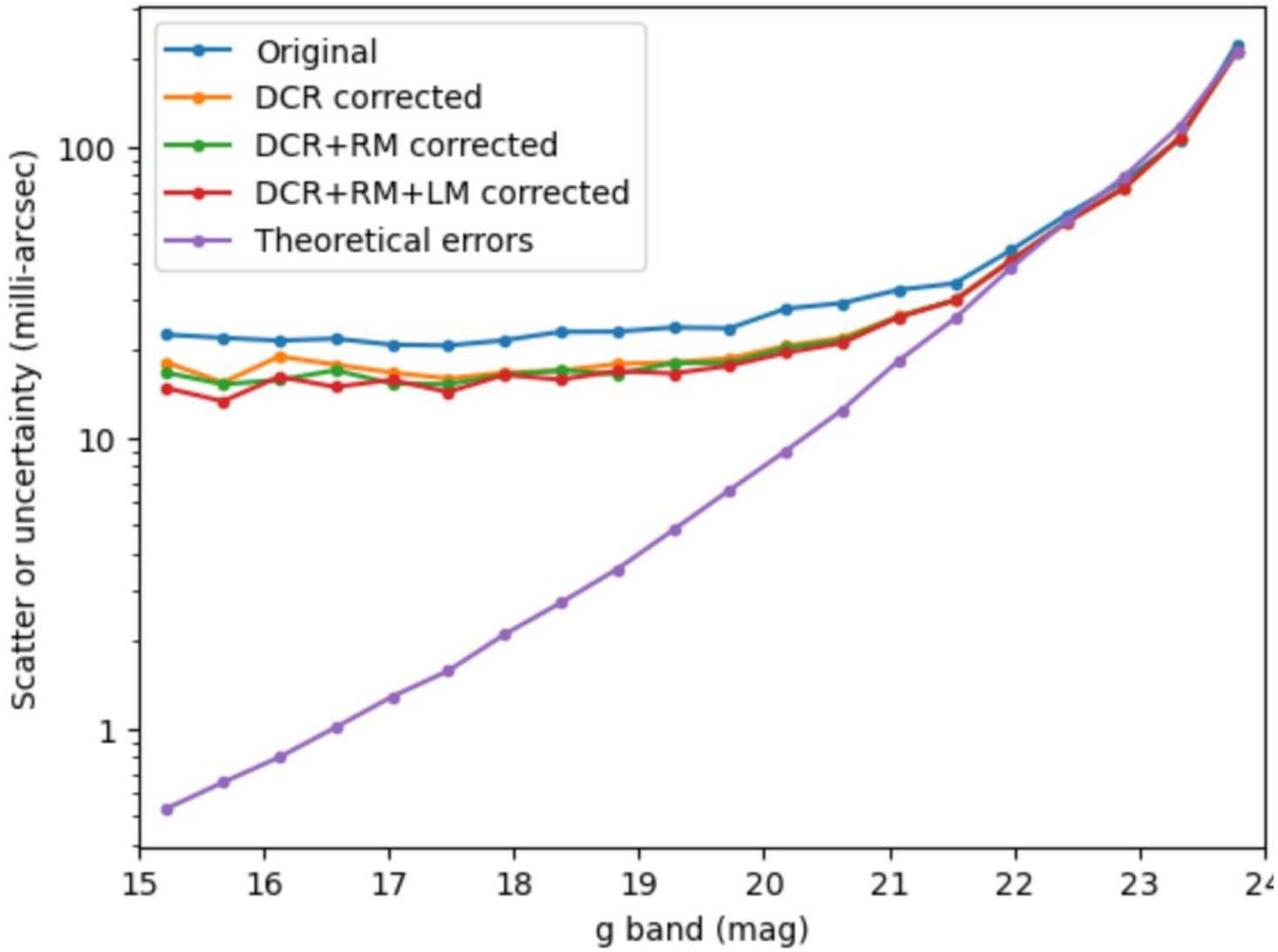
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Late



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Error Analysis



Summary

- Why we need proper motions
  - Learn about MW + galaxies
- Gaia vs. DESI Legacy
  - Legacy can see fainter stars
- Systematic errors
  - Differential Chromatic Refraction (DCR)
  - CCD-specific corrections
  - Lateral colour
- Results and error analysis

- **In Progress:**
  - Apply corrections to all data and produce catalogues